

Coamenability versus Relative Amenability for Coideals: A Scoped Later Answer

Literature status for arXiv:2003.04384

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Status: literature already answered in two broad regimes; the unrestricted question remains open. Anderson–Sackaney and Khosravi, arXiv:2311.14587, prove the desired coamenability/relative-amenability equivalence when the ambient discrete quantum group is unimodular or exact. In arbitrary ambient groups they prove one implication and obtain a full equivalence after strengthening relative amenability to equivariant injectivity.

The source question

Question 1.3 of Anderson–Sackaney’s source paper asks the following. Let $\mathbb{G}$ be a discrete quantum group and let $\widehat{\mathbb{H}}$ be a closed quantum subgroup of $\widehat{\mathbb{G}}$. Is

$$L^\infty(\widehat{\mathbb{G}}/\widehat{\mathbb{H}}) \text{ coamenable} \iff \ell^\infty(\mathbb{H}) \text{ relatively amenable?}$$

The next sentence gives the compact-quasi-subgroup version: for an idempotent state ω , is the compact coideal N_ω coamenable exactly when its codual coideal $\widetilde{N}_{P_\omega}$ is relatively amenable? The source explicitly says that it only makes progress on this version [1]. The question and extension occur on source PDF page 4.

The later theorem

The supporting paper uses the paired notation

$$\widehat{\mathbb{X}} \backslash \widehat{\mathbb{G}} \quad \text{and} \quad \ell^\infty(\mathbb{X})$$

for a compact-quasi-subgroup coideal and its codual. This is the same coideal–codual formulation as the source’s N_ω and $\widetilde{N}_{P_\omega}$; quotient-side symbols depend on the chosen left/right convention.

Its two main implications are:

$$\begin{aligned} \widehat{\mathbb{X}} \backslash \widehat{\mathbb{G}} \text{ coamenable} &\implies \ell^\infty(\mathbb{X}) \text{ is } \mathbb{G}\text{-injective} && \text{(Theorem 3.2, PDF page 12),} \\ \ell^\infty(\mathbb{X}) \text{ is } \mathbb{G}\text{-injective} &\implies \widehat{\mathbb{X}} \backslash \widehat{\mathbb{G}} \text{ coamenable} && \text{(Theorem 6.5, PDF page 25).} \end{aligned}$$

Thus, without any unimodularity or exactness assumption,

$$\widehat{\mathbb{X}} \backslash \widehat{\mathbb{G}} \text{ is coamenable} \iff \ell^\infty(\mathbb{X}) \text{ is } \mathbb{G}\text{-injective.}$$

Since $\mathbb{G}$ -injectivity implies relative amenability, the forward implication asked for in the source follows for every compact quasi-subgroup. These statements are author-explicit: the abstract and introduction of the supporting paper state this coideal/codual equivalence directly [2].

When the requested equivalence is complete

Remark 6.6 on supporting PDF page 27 proves the missing implication

$$\ell^\infty(\mathbb{X}) \text{ relatively amenable} \implies \widehat{\mathbb{X}} \backslash \widehat{\mathbb{G}} \text{ coamenable}$$

under either of the following assumptions:

1. $\mathbb{G}$ is unimodular;
2. $C(\widehat{\mathbb{X}} \backslash \widehat{\mathbb{G}})$ is exact.

In particular, the second condition holds whenever $\mathbb{G}$ is exact, because $C(\widehat{\mathbb{X}} \backslash \widehat{\mathbb{G}})$ is a C^* -subalgebra of $C(\widehat{\mathbb{G}})$. Combining Remark 6.6 with Theorem 3.2 therefore yields the exact equivalence asked for in the compact-quasi-subgroup extension whenever $\mathbb{G}$ is unimodular or exact. Coideals of quotient type are included in this class, so the corresponding closed-quantum-subgroup question has the same scoped affirmative answer.

The proof mechanisms also explain the two hypotheses. In the unimodular case, Theorem 6.2 upgrades relative amenability to $\mathbb{G}$ -injectivity. In the exact case, the proof of Theorem 6.5 can use a nuclear inclusion supplied by injectivity of the von Neumann coideal in place of nuclearity of the reduced C^* -coideal.

What is not settled

The later paper does not prove that relative amenability implies $\mathbb{G}$ -injectivity for an arbitrary non-unimodular, non-exact discrete quantum group, nor does it prove the converse implication in the source question without one of the above hypotheses. Its unrestricted exact replacement is $\mathbb{G}$ -injectivity, which is strictly stronger at the level of the definitions. Accordingly, this packet records a substantial scoped literature answer, not a full resolution of Question 1.3 in complete generality.

Verification and provenance

The source question and compact-quasi-subgroup extension were checked on source PDF page 4. In the supporting paper, the summary equivalence was checked in the abstract (page 1) and introduction (page 2), the universal forward implication in Theorem 3.2 (page 12), the injectivity converse in Theorem 6.5 (page 25), and the unimodular/exact upgrade in Remark 6.6 (page 27). The supporting paper is coauthored by the source author and cites the source for the coideal framework. No novelty claim is made here.

References

- [1] B. Anderson–Sackaney, *On amenable and coamenable coideals*, J. Funct. Anal. **286** (2024), no. 10, 110384; arXiv:2003.04384.
- [2] B. Anderson–Sackaney and F. Khosravi, *Topological boundaries of representations and coideals*, arXiv:2311.14587v2 (2024).